

Reconciling the Delphes $\tau_h\tau_h$ channel with CMS NanoAOD

A cutflow diagnostic, seven corrections, and a decision to take

Jingjing Pan

August 11, 2026

Abstract

The tuned Delphes $HH \rightarrow b\bar{b}\tau^+\tau^-$ sample initially selected $\sim 5\times$ *fewer* $\tau_h\tau_h$ events than the CMS NanoAOD reference, and its $m_{\tau\tau}$ peaked ~ 25 GeV high. A five-stage cutflow localised the yield deficit to a single stage — the FastMTT di- τ mass — and the cause turned out not to be a detector or tuning effect but a representational one: **in Delphes a τ_h is an AK4 jet, whereas in CMS it is an HPS τ_h object**. Nearly every discrepancy we found traces to that one difference, expressed in a different component of the four-vector or in a different reference used to tune it. Seven corrections are described, each anchored to the CMS NanoAOD and applied downstream so that no Delphes re-production is required. The $\tau_h\tau_h$ yield deficit is closed — the cutflow now agrees with CMS at *every* stage, ratio 1.01 — and the $m_{\tau\tau}$ peak offset has fallen from ~ 25 to ~ 7 GeV; seven of the ten NSBI input features agree with CMS within statistics. One discrepancy remains open, in $\Delta R_{\tau\tau}$. A methodological lesson is recorded: three of our own errors came from confusing generator *copies* with distinct particles. Section 6 sets out a decision we would like τ expertise on: whether to keep correcting the jet-based τ_h , or to rebuild it from the Delphes particle-flow candidates.

1 Motivation and summary

The Delphes fast-simulation sample must reproduce the CMS $HH \rightarrow b\bar{b}\tau\tau$ Run-3 baseline closely enough to support a phenomenological unbinned NSBI κ_λ measurement. The $b\bar{b}$ side and the κ_λ -carrying observable m_{HH} agreed from early on; the $\tau\tau$ side did not, in two ways that resisted object-level tuning: a factor ~ 5 yield deficit and a ~ 25 GeV shift in $m_{\tau\tau}$.

Table 1 is the summary. Each row is one correction, all of them downstream of the Delphes card (the production is unchanged), and each was adopted only after a measurement identified it. Two further fixes — described in Section 4 — were to the *diagnostic* rather than to the simulation, and are kept separate for that reason.

#	correction	Delphes N	Delphes/CMS	$m_{\tau\tau}$ peak offset
0	(starting point, symmetric selection)	289	0.20	+25
1	τ_h visible <i>mass</i> from the anchor	2590	1.81	+25
2	hadronic vs leptonic gen τ	2100	1.47	+25
3	common τ -candidate acceptance	1729	1.30	+25
4	τ energy scale anchored to CMS	1721	1.30	+25
5	p_T^{miss} smearing (pileup)	1721	1.30	+12
6	visible- τ energy reference	1660	1.25	$\sim +7$
7	energy scale applied <i>before</i> the tag draws	1569	1.18	$\sim +7$

Table 1: Cumulative effect of the seven corrections, at $\kappa_\lambda = 0$ from 2×10^4 generated events (CMS reference: 1431 events, falling to 1328 once the common τ acceptance is applied to both sides). Peak offsets are read off the normalised histograms and are approximate. These are the *analysis*-level yields (the overlay, which requires no generator matching); the gen-matched cutflow of Table 3 closes further, to 1.01.

2 The diagnostic

Both samples are walked through the same five stages; a stage whose *conditional* efficiency differs between the two columns is where a difference is created.

#	stage	Delphes	CMS NanoAOD
1	gen	≥ 2 visible hadronic gen τ	≥ 2 GenVisTau
2	reco	≥ 2 accepted jets matched to those	≥ 2 Tau objects matched
3	ID	≥ 2 passing the re-tagged TauTag	≥ 2 passing DeepTau v2.5 VSjet Medium
4	jets	≥ 2 jets surviving the overlap removal	identical
5	$m_{\tau\tau}$	FastMTT mass finite and > 20 GeV	identical

Table 2: Cutflow stages. τ candidates $p_T > 20$ GeV, $|\eta| < 2.3$; jets $p_T > 20$ GeV, $|\eta| < 2.4$; matching and overlap removal $\Delta R < 0.4$.

stage	initial			after correction 1			final		
	$\varepsilon_{\text{step}}$	D/CMS	$\Delta\varepsilon$	$\varepsilon_{\text{step}}$	D/CMS	$\Delta\varepsilon$	$\varepsilon_{\text{step}}$	D/CMS	$\Delta\varepsilon$
1 gen visible τ_h	0.499	/ 0.235	2.12	0.499	/ 0.235	2.12	0.229	/ 0.235	0.97
2 reco τ_h object	0.635	/ 0.855	0.74	0.635	/ 0.855	0.74	0.946	/ 0.855	1.11
3 τ_h ID	0.370	/ 0.386	0.96	0.370	/ 0.386	0.96	0.351	/ 0.386	0.91
4 cleaned jets	0.865	/ 0.885	0.98	0.865	/ 0.885	0.98	0.885	/ 0.885	1.00
5 FastMTT $m_{\tau\tau}$	0.120	/ 0.991	0.12	1.000	/ 0.991	1.01	1.000	/ 0.991	1.01
CMS/Delphes yield	5.60			0.67			1.01		

Table 3: Cutflow at $\kappa_\lambda = 1$, 2×10^4 events. $\Delta\varepsilon$ is the Delphes/CMS ratio of the per-stage efficiency. Stage 5 alone carried the entire initial deficit. In the final column every stage agrees with CMS individually.

Two readings matter. First, **stage 5 alone** carried the original deficit: Delphes retained 12% where CMS retained 99%, and through stage 4 Delphes was in fact *ahead* in cumulative efficiency. Second, in the final column every stage agrees individually — which is a stronger statement than the total agreeing, and was reached only after the two diagnostic fixes of Section 4.

The stage-2/3 split deserves a word, since 1.11 and 0.91 might look like two errors cancelling. They are not: the product is $0.946 \times 0.351 = 0.332$ against $0.855 \times 0.386 = 0.330$, agreeing to 0.6%, and the split differs *by construction*. The anchor’s τ_h efficiency is measured as **GenVisTau**→DeepTau-Medium **Tau**, so it already contains CMS’s HPS *reconstruction* inefficiency ($\sim 15\%$). On the Delphes side a τ almost always yields a jet (stage 2 $\simeq 0.95$) and the map then applies the combined reconstruction×identification at stage 3. The product is the quantity the map guarantees, and it is what the comparison should test.

3 One root cause, several components

A CMS τ_h is an HPS object built from a narrow set of decay products. A Delphes τ_h is an anti- k_T $R = 0.4$ jet that happens to carry **TauTag**. The two therefore differ in *every* component of the four-vector, and additionally in the generator-level reference one would naturally use to tune each component:

- **mass** — the jet mass is $\mathcal{O}(5\text{--}20)$ GeV; a τ_h visible mass is bounded by $m_\tau = 1.777$ GeV (correction 1);
- **energy** — the jet carries the underlying event inside the cone, so it is $\sim 17\%$ harder than the visible τ at low p_T (corrections 4 and 6);

- **reference** — a Delphes **GenJet** is also a jet and also contains the underlying event, so it is *not* the analogue of **GenVisTau** (correction 6);
- **identity** — “matched to a gen τ ” does not mean “hadronic” (correction 2), and the two data tiers expose τ candidates with different intrinsic thresholds (correction 3).

Only correction 5 (p_T^{miss}) is unrelated to this; it follows from the card running without pileup by design.

3.1 Correction 1: the τ_h visible mass, and why it must be sampled

The covariance-free FastMTT estimator maximises, per event, a likelihood over the visible energy fractions $x_1, x_2 \in (0, 1]$, with $m_{\tau\tau} = m_{\text{vis}}/\sqrt{x_1 x_2}$ and a hadronic decay prior

$$\mathcal{L}_{\text{dec}}(x) = \begin{cases} 1, & x_{\min} \leq x \leq 1, \\ 0, & \text{otherwise,} \end{cases} \quad x_{\min} \equiv (m_{\text{vis}}/m_\tau)^2. \quad (1)$$

If $m_{\text{vis}} > m_\tau$ then $x_{\min} > 1$, the admissible interval is empty, the likelihood vanishes identically and the estimator returns NaN. A Delphes τ -jet carrying its jet mass therefore admits *no solution at all*: $\sim 89\%$ of $\tau_h \tau_h$ pairs were silently lost, which was the whole of the stage-5 collapse.

The correction measures the CMS τ_h visible mass on the anchor and assigns it to τ -tagged Delphes jets. Crucially it must be **sampled, not averaged**. An initial implementation assigned the per- p_T median; that is wrong because via Eq. (1) the mass is a one-sided *lower bound* on x , not a smearing. Its contribution to the numerator m_{vis} is negligible ($m^2 \sim 1.6$ against $m_{\text{vis}}^2 \sim 3600 \text{ GeV}^2$), so a single value *relaxes* a real constraint for every leg whose true mass lies above it. For a 3-prong a_1 leg ($m_{\text{vis}} = 1.26$, floor $x_{\min} = 0.503$) given the median 0.77 ($x_{\min} = 0.188$), with p_T^{miss} favouring $x \simeq 0.25$, the fit moves from $x = 0.525$ to $x = 0.250$ and $m_{\tau\tau}$ from 224 to 470 GeV. Over a realistic decay-mode mixture the median assignment inflates the $m_{\tau\tau}$ width by $\sim 60\%$ and the mean by $\sim 9\%$ while leaving the *mode* unchanged — so the $Z \rightarrow \tau\tau$ candle, which gates the mode, would not have caught it. The implementation therefore stores the anchor’s per- p_T mass *quantiles* and draws per τ -jet from them.

3.2 Corrections 2–3: identity and acceptance

The τ_h efficiency map is measured on the anchor as **GenVisTau** $\rightarrow$ DeepTau-Medium **Tau**, and **GenVisTau** exists only for hadronic decays. Treating every gen τ as genuine handed a leptonic τ ’s jet that efficiency (~ 0.5) instead of the jet $\rightarrow \tau_h$ fake rate (~ 0.004), fabricating $\tau_h \tau_h$ events out of $\tau_h \tau_\ell$ and $\tau_\ell \tau_\ell$ ones. A gen τ is now classified by looking for a status-1 e/μ whose first non-self-copy ancestor is a τ , using the generator mother links only.

Separately, the two tiers expose τ candidates with different intrinsic floors: a Delphes τ_h is a jet and so exists down to the card’s **JetPMin** of 15 GeV and out to $|\eta| < 2.5$, while the CMS **Tau** collection begins near 18–20 GeV. A common acceptance ($p_T > 20$, $|\eta| < 2.3$, the analysis values) is now applied to both sides before the pair is chosen.

3.3 Corrections 4 and 6: the τ energy scale and its reference

The energy-scale map was originally derived by inverting the Delphes response to unity against its own **GenJet** — a *self*-comparison, where every other map in the framework is a comparison *to CMS*. Correction 4 replaced it with the anchor ratio, **tau_escale** = $\mathcal{R}_{\text{CMS}}/\mathcal{R}_{\text{Delphes}}$, in the same form already used for the lepton scale factors.

That alone changed almost nothing (1–3%), and the reason is instructive: the two responses were still measured against *different* references. The CMS response is **Tau/GenVisTau** — clean — while the Delphes response was τ -jet/**GenJet**, and a **GenJet** is an $R = 0.4$ cluster that still contains

the underlying event (only *pileup* is absent from the card). Correction 6 builds a genuine visible gen τ on the Delphes side using the identity that a hadronic τ decay has exactly one neutrino,

$$p_{\text{vis}} = p_{\tau} - p_{\nu_{\tau}}, \quad (2)$$

with the ν_{τ} located through the same mother-link walk (no generator daughter links are needed). Profiled against *that*, the Delphes τ -jet is **17% harder than the visible τ at 25 GeV**, falling to $\sim 2\%$ at 250 GeV — the shape expected of a roughly fixed underlying-event offset. The resulting map is `tau_escale` = [0.85, 0.88, 0.90, 0.91, 0.93, 0.94, 0.94, 0.95], against [0.91, ..., 0.96] before.

3.4 Correction 5: p_T^{miss} resolution (pileup)

The card runs without pileup by construction, and its header states that “pileup enters through the tuning maps”. That map had never been built. Measured with the same estimator on both sides, the per-component p_T^{miss} resolution is 16.5 GeV on Delphes against 32.6 GeV on the CMS anchor — a factor of two, and the largest single mismatch anywhere in the map set. Gaussian noise of the quadrature difference (28.1 GeV) is added per component. Since p_T^{miss} is what FastMTT fits the energy fractions against, this is the correction that moved $m_{\tau\tau}$ most: the peak offset fell from ~ 25 to ~ 12 GeV and the distribution broadened from far-too-narrow into the right range.

4 Generator copies, and two fixes to the diagnostic

Three of our own errors had a single cause, and it is worth recording because it is specific to reading an unpruned generator record. Delphes writes `allParticles` — the full Pythia history — so each physical τ appears several times (status 23 $\rightarrow$ 22 $\rightarrow$ 2...). CMS NanoAOD `GenPart` is pruned and carries essentially one entry per τ ; measured against `GenVisTau` our construction gives an object ratio of 0.995 there, so the issue is invisible on the anchor and appears only on the Delphes side.

The rule that emerged:

Anything that **counts** generator objects must collapse copies first. Anything that **matches** them geometrically need not — copies are collinear, so they collapse automatically under a ΔR or nearest-neighbour match.

That distinction explains why the energy-scale maps, the re-tagging and the overlay were never affected (all matching), while the cutflow’s stage-1 *count* was.

Fix A: the stage-1 reference. Stage 1 originally counted `GenJets` matched to a hadronic τ . But a `GenJet` is an $R = 0.4$ cluster carrying the underlying event — $\sim 17\%$ harder than the visible τ at low p_T — so a $p_T > 20$ GeV cut applied to it admitted $\sim 8\%$ more τ per leg than the CMS side, i.e. $\sim 17\%$ more pairs. This is the same reference error that correction 6 removed from the energy scale, in a second place. Stage 1 now uses the neutrino-subtracted τ directly.

Fix B: copy collapse. With Fix A alone, stage 1 *rose* to $\Delta\varepsilon = 1.48$, because the visible- τ construction returned one entry per τ *copy* — so “ ≥ 2 visible τ ” could be satisfied by two copies of a single τ . Only the last copy is now kept (a τ that no other τ lists as its mother). Stage 1 then lands at 0.97.

A rejected “improvement”. We also replaced the geometric leptonic- τ veto (is a τ -descended e/μ within ΔR of the τ ?) with an exact one based on the mother links (*which* τ did this lepton descend from?). Scored against CMS `GenVisTau` on 2×10^5 anchor events, the “exact” version is decisively worse:

veto	objects	efficiency	purity	stage-1 ratio
geometric (kept)	$0.995\times$	0.991	0.996	0.989
descent (rejected)	$1.186\times$	0.986	0.831	1.460

because it vetoes only the single copy the lepton’s chain points at, leaving the others as fake hadronic τ . The geometric test is accidentally copy-robust: every copy sits at the same η, ϕ , so all are removed together. It is kept.

5 What is still open

The gen-matched yield is closed (Table 3). What remains is one feature.

The $\Delta R_{\tau\tau}$ shoulder. Delphes carries an excess at $\Delta R_{\tau\tau} \simeq 2.2\text{--}3.2$ that CMS does not, and it has survived every correction above. A dedicated diagnostic separates the two possible causes by comparing, on each side, the *generator-level* ΔR between the two visible hadronic τ and the selection efficiency against it:

- the gen $\Delta R_{\tau\tau}$ *spectra agree*, so the samples are the same physics and no generator-level explanation is available;
- the *efficiencies* genuinely differ: Delphes selects 0.22–0.23 of events at gen $\Delta R > 2.5$ where CMS selects 0.15–0.17 ($\sim 3\sigma$), with a weaker ($\sim 2\sigma$) hint that Delphes also *loses* pairs below $\Delta R < 1$.

Both push the normalised distribution rightward. The low- ΔR hint is what one would expect if collimated pairs merge into a single $R = 0.4$ jet, but CMS HPS also seeds from AK4 jets, so the merging scales are similar and this cannot be the whole story. The analysis-level (non-gen-matched) yield retains a $1.18\times$ excess, plausibly the same events.

A possible route. The card writes the Delphes particle-flow candidates (the three EFlow collections: tracks, photons and neutral hadrons), and its header lists “ τ redefinition” as a reason for keeping them. Rebuilding τ_h HPS-style from those — a narrow signal cone rather than an $R = 0.4$ jet — would address the root cause directly and would *retire* corrections 1, 4 and 6 rather than adding an eighth. It would not reproduce DeepTau, so the efficiency and fake-rate maps would remain. Two caveats: Delphes neutral candidates are calorimeter towers, so π^0 reconstruction (which sets the mass and the decay mode) would be approximate; and since CMS also seeds from jets, it is not clear this fixes the ΔR shoulder.

A single candidate explains both, and is our current working hypothesis. The downstream corrections are applied in the order (b-tag, τ_h tag, energy scale, mass), so the τ_h efficiency is drawn at the *raw* jet p_T — which correction 6 shows to be $\sim 17\%$ too high at low p_T . The efficiency rises steeply there ($0.39 \rightarrow 0.52$ between 25 and 35 GeV), so evaluating it $\sim 17\%$ too far to the right produces a systematic over-efficiency for soft τ : too many reconstructed pairs, concentrated exactly in the low- p_T , large- $\Delta R_{\tau\tau}$ population. Applying the energy scale *first* would make every subsequent map evaluate at a CMS-like p_T . This is falsifiable: if the hypothesis is right the excess should vanish preferentially at low τ p_T and take the shoulder with it; if the shoulder survives a corrected low- p_T efficiency it is genuinely angular, and the τ direction (jet axis versus HPS axis) is the next candidate.

Hypotheses tested and rejected. For the record, three explanations for the $m_{\tau\tau}$ shift were proposed and excluded by direct measurement: (i) *fake τ_h* — the gen-matched/fake split gives fake fractions of 5.7% (Delphes) against 5.0% (CMS), essentially equal, with both shape discrepancies living entirely in the gen-matched component; (ii) *τ -pair selection* — making the selection fully symmetric moved $\Delta R_{\tau\tau}$ but left $m_{\tau\tau}$ untouched; (iii) *a large CMS τ energy miscalibration* — the anchor response is 0.97–0.99, so CMS τ_h are calibrated to the visible τ to within a few percent.

Not addressed here. Pileup also affects the jet energy *resolution* (only the scale is corrected) and produces pileup jets that Delphes cannot make at all. Both act on the $b\bar{b}$ side, which currently agrees well with CMS, and the second cannot be fixed downstream in any case.

6 Decision: keep correcting the jet, or rebuild the τ_h ?

Every correction above treats the symptom of one fact: a Delphes τ_h is an AK4 jet. The alternative is to stop correcting it and instead *rebuild* τ_h from the Delphes particle-flow candidates, which the card already writes (the three EFlow collections — tracks, photons, neutral hadrons — kept, per the card header, precisely for “ τ redefinition”). We would value τ -expert judgement before committing.

What rebuilding would involve. An HPS-like construction: seed from a jet, collect charged hadrons and photons in a narrow signal cone (fixed $\Delta R \simeq 0.1$, or the p_T -dependent $3.0/p_T$ bounded at 0.05–0.1), form the standard decay-mode hypotheses (1-prong, 1-prong+ π^0 , 3-prong), take the τ_h four-vector as the sum of signal-cone constituents, and use an isolation annulus for a handle on fakes.

What it would gain. The mass, energy and direction would all be correct *by construction* rather than by correction, and corrections 1, 4 and 6 — the visible-mass map, the CMS-anchored energy scale and the visible- τ reference — would be *retired* rather than joined by an eighth. For a phenomenological paper this is a materially stronger statement than a jet-plus-corrections chain, and it removes a stated fast-simulation limitation rather than documenting it.

What it would not do. It cannot reproduce DeepTau, so the identification efficiency and the jet $\rightarrow \tau_h$ fake rate would remain anchor-measured maps (these already work). And it is *not* obviously the cure for the open $\Delta R_{\tau\tau}$ discrepancy: CMS HPS also seeds from AK4 jets, so the two merging scales are similar and a narrow signal cone may not change which pairs are resolved.

Cost, honestly. The implementation is a contained piece of work — hours to days of coding plus validation — and is *not* the critical path; running the full 5×10^5 -event samples through the ladder and training the NSBI is the multi-day item. So the two are not really in competition for schedule, only for attention.

The Delphes-specific caveat we most want checked. Delphes neutral particle-flow candidates are *calorimeter towers* ($\sim 0.02 \times 0.02$ in the barrel), not clustered PF objects. Since the π^0 content is what sets both the visible mass and the decay-mode assignment, an HPS-like reconstruction on Delphes would be approximate in exactly the place that matters most.

Questions for τ expertise.

1. Is tower-granular neutral reconstruction adequate for a useful decay-mode assignment, or does it degrade the π^0 content enough that the rebuilt mass is no better than the anchor-sampled one we already have?
2. Signal cone: fixed 0.1, or p_T -dependent? Does the choice matter at the p_T range that dominates $HH \rightarrow b\bar{b}\tau\tau$ (20–60 GeV)?
3. Seeding: jets (as CMS HPS does) or direct clustering of particle-flow candidates? Would either change the efficiency for collimated pairs, $\Delta R_{\tau\tau} \lesssim 1$?

4. Is the observed pattern — Delphes over-efficient at gen $\Delta R_{\tau\tau} > 2.5$ ($\sim 3\sigma$), possibly under-efficient below 1 ($\sim 2\sigma$) — something you recognise as a jet-vs-HPS effect at all, or should we look elsewhere?
5. Is a jet-based τ_h with anchor-measured corrections defensible in a phenomenological paper, or is rebuilding the expected standard?

7 Points for cross-check

1. **Is the framing accepted?** The claim is that essentially the whole τ -side discrepancy follows from representing a τ_h as an AK4 jet, and that the remedy is to correct each component of that four-vector against a CMS-anchored, like-for-like reference.
2. **Is quantile sampling of the visible mass sufficient?** The spread is reproduced, but the drawn mass is independent of the τ energy and of the partner leg. Would sampling a decay mode per τ be preferable, so that mass, prong multiplicity and energy sharing stay correlated?
3. **Is isotropic p_T^{miss} smearing adequate?** Real p_T^{miss} resolution is anisotropic and scales with ΣE_T ; ours is round and flat. Delphes $m_{\tau\tau}$ is now marginally *broader* than CMS, which is what one would expect from over-simplified noise.
4. **Is the $\Delta R_{\tau\tau}$ shoulder worth pursuing further**, given that it is the last open item and the ten NSBI features otherwise agree? The candidate remedy (rebuilding τ_h from particle-flow candidates) is the largest single piece of work in this programme, and it is not certain to fix it.
5. **Residual normalisation.** The gen-matched cutflow closes to 1.01, but the analysis-level selection retains $1.18\times$. Is that acceptable for an unbinned NSBI study, where the training uses shapes and the normalisation is fitted?

Reproducing

All corrections live in `delphes-pipeline` and are applied downstream of the Delphes card; the existing production is unchanged. Re-derive the maps and remake the comparison with

```
python -m delphes_pipeline.tuning.derive_maps --config config.v1.yml \
    --output cards/tuning/maps_v1.json
python scripts/tauh_cutflow.py --config config.v1.yml [...] # the diagnostic
python scripts/nsbi_overlay.py --config config.v1.yml [...] \
    --tautau-only --diagnostics
```

The cutflow’s unit tests inject a loss at one known stage at a time and assert it is attributed to that stage and no other; the overlay’s tests assert that the same physical object is accepted or rejected identically on both sides.